
Target-Speaker Speech-to-Text in Overlapping Speech

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Abstract

Modern speech-to-text systems achieve low word error rates on clean, single-speaker audio but degrade sharply when speakers overlap. We present a target-speaker speech-to-text system that, given a short enrollment clip of the desired speaker and a multi-speaker audio mixture, transcribes only the target speaker. Our pipeline combines a frozen pretrained speaker verification model (ECAPA-TDNN) to extract a speaker embedding, a trained speaker-conditioned masking network that isolates the target speaker's spectrogram, and a frozen pretrained ASR model (Whisper) for transcription. We evaluate on synthetic two-speaker mixtures from LibriSpeech at varying signal-to-interferer ratios and overlap ratios, measuring word error rate and character error rate. We show that overlapping speech degrades ASR performance across languages, and present our extraction-based approach to mitigate this degradation.

Attestation of Teamwork

Jibran Iqbal Shah: Designed and trained the speaker-conditioned extraction network, including architecture design, loss function development (power-law compression, hybrid oracle mask loss), hyperparameter tuning, and all related code and writeup.

Yamoah Frimpong Attafuah: Fine-tuned Whisper-small on the WAXAL Akan dataset for Twi speech recognition, set up the WER evaluation scripts and notebooks, and benchmarked both the baseline Whisper and fine-tuned Whisper models across varying overlap ratios and SIR levels for English and Twi. Wrote the code and writeup related to these contributions.

1 Introduction

Modern speech-to-text systems perform well on clean, single-speaker audio but degrade sharply when speakers overlap [Watanabe et al., 2020, Kanda et al., 2020]. This is a common occurrence in meetings, court proceedings, and media such as films or talk shows. Rather than attempting full diarization and separation, target-speaker transcription offers a more focused alternative: given a short enrollment clip, transcribe only the speaker of interest.

Additionally, the rise of full-duplex speech-to-speech models is increasing the demand for labelled transcriptions of natural conversations with interruptions [Défossez et al., 2024, Roy et al., 2026]. A reliable target-speaker system could automatically generate such labels by isolating each participant and producing timed transcripts, even through overlapping speech.

The objective of this project is to develop a target-speaker speech-to-text system that takes (i) a short enrollment utterance of the target speaker and (ii) a multi-speaker mixture, and outputs only the target speaker’s transcript. We test across English and Twi to examine whether the degradation pattern is language-agnostic.

2 Preliminaries and Problem Formulation

Let $\mathbf{x}_{\text{mix}} \in \mathbb{R}^T$ be a single-channel audio mixture containing overlapping speech from $N \geq 2$ speakers. We are also given a short enrollment utterance $\mathbf{x}_{\text{enroll}} \in \mathbb{R}^{T_e}$ of the target speaker s^* . The goal is to produce a transcript $\hat{y}$ that matches the ground-truth transcript y^* of speaker s^* within the mixture, as measured by word error rate $\text{WER} = (S + D + I)/N_{\text{ref}}$, where S, D, I are the number of substitutions, deletions, and insertions and N_{ref} is the number of words in the reference transcript. Note that WER can exceed 100% when the model produces more insertions than there are reference words.

The Short-Time Fourier Transform (STFT) converts a waveform into a two-dimensional complex-valued matrix $\mathbf{X} \in \mathbb{C}^{F \times T}$, where each entry encodes the amplitude and phase of frequency bin f during time frame t . The magnitude spectrogram $|\mathbf{X}| \in \mathbb{R}^{F \times T}$ retains full linear frequency resolution, which is needed to invert back to a waveform after masking via the inverse STFT (iSTFT). We use standard parameters for 16 kHz speech: 25 ms window (400 samples), 10 ms hop (160 samples), FFT size $N_{\text{fft}} = 512$, giving $F = 257$ frequency bins linearly spaced from 0–8000 Hz.

3 Design

Our pipeline consists of three stages, illustrated in Figure 1:

1. Speaker embedding extraction: A frozen pretrained ECAPA-TDNN [Desplanques et al., 2020] encodes the enrollment utterance $\mathbf{x}_{\text{enroll}}$ into a speaker embedding $\mathbf{e} \in \mathbb{R}^{192}$.
2. Speaker-conditioned extraction: A trained masking network receives the mixture magnitude spectrogram $|\mathbf{X}_{\text{mix}}|$ and the speaker embedding $\mathbf{e}$, and predicts a soft mask $\mathbf{M} \in [0, 1]^{F \times T}$ that isolates the target speaker. The estimated target spectrogram is $|\hat{\mathbf{X}}_{\text{target}}| = \mathbf{M} \odot |\mathbf{X}_{\text{mix}}|$, and the waveform is reconstructed via iSTFT using the original mixture phase.
3. ASR decoding: A frozen pretrained Whisper model transcribes the extracted waveform.

Only the extraction network (stage 2) is trained; the speaker encoder and ASR model remain frozen throughout.

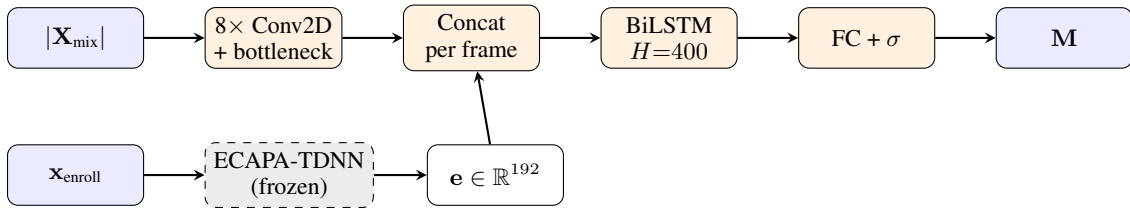


Figure 1: Extraction network architecture. The magnitude spectrogram is encoded by eight 2D convolutional layers, conditioned on the speaker embedding via concatenation, and processed by a BiLSTM to predict a soft mask $\mathbf{M} \in [0, 1]^{F \times T}$. Dashed box denotes a frozen pretrained model.

4 Methodology

4.1 Speaker embedding and extraction network architecture

We use a pretrained ECAPA-TDNN [Desplanques et al., 2020] (speechbrain/spkrec-ecapa-voxceleb) to extract a 192-dimensional speaker embedding from the enrollment utterance. The model achieves 0.80% equal error rate on VoxCeleb1 and is kept frozen throughout.

The extraction network follows the VoiceFilter design [Wang et al., 2019] and has 9.3M trainable parameters. The magnitude spectrogram is treated as a single-channel 2D image and processed by eight Conv2D layers with layer normalization and ReLU. The first two layers use separable kernels (frequency-only and time-only), layers 3–7 use joint (5, 5) kernels with time-only dilation (factors 1, 2, 4, 8, 16) giving a temporal receptive field of 31 frames (≈ 310 ms), and the final layer is a 1×1 bottleneck reducing from 64 to 8 channels.

The CNN output is flattened per time frame and concatenated with the speaker embedding $\mathbf{e} \in \mathbb{R}^{192}$. A single-layer BiLSTM ($H=400$ per direction) processes the sequence, and two FC layers with ReLU and sigmoid produce the mask $\mathbf{M}_t \in [0, 1]^F$ at each frame. We use a BiLSTM rather than a unidirectional LSTM because overlap regions are locally ambiguous and both past and future context help disambiguate which spectral energy belongs to the target speaker.

4.2 Training objective and data pipeline

We train with a power-law compressed reconstruction loss following Wilson et al. [Wilson et al., 2018], who show that element-wise power-law compression of spectrograms before computing MSE improves separation quality. We use an exponent of $p = 0.3$, which we confirmed through a short ablation over $p \in \{0.1, 0.2, 0.3, 0.5, 1.0\}$:

$$\mathcal{L}_{\text{recon}} = \frac{1}{FT} \left\| (\mathbf{M} \odot |\mathbf{X}_{\text{mix}}| + \epsilon)^p - (|\mathbf{X}_{\text{target}}| + \epsilon)^p \right\|_F^2$$

where $\epsilon = 10^{-8}$. The same compression is applied to the CNN input to normalize its dynamic range; the mask is still applied to raw magnitudes for reconstruction.

We also experimented with a hybrid loss that adds direct supervision on the oracle mask. Since the mask should satisfy $\mathbf{M} \odot |\mathbf{X}_{\text{mix}}| = |\mathbf{X}_{\text{target}}|$, dividing both sides by $|\mathbf{X}_{\text{mix}}|$ gives the ideal mask $\mathbf{M}_{f,t}^* = \text{clamp}(|\mathbf{X}_{\text{target}}|_{f,t} / (|\mathbf{X}_{\text{mix}}|_{f,t} + \epsilon), 0, 1)$, clamped to $[0, 1]$ since the sigmoid output cannot exceed this range:

$$\mathcal{L} = \alpha \cdot \|\mathbf{M} - \mathbf{M}^*\|_F^2 + (1 - \alpha) \cdot \mathcal{L}_{\text{recon}}$$

This hybrid approach is inspired by Wilson et al. [Wilson et al., 2018], who explore tradeoffs between mask-domain and signal-domain losses. We found $\alpha = 0.1$ to best balance optimizing the oracle mask loss and the reconstruction loss, through a sweep over $\alpha \in \{0, 0.05, 0.1, 0.2, 0.3, 1.0\}$.

We use LibriSpeech (dev-clean + test-clean, 16 kHz mono) for English and WAXAL Akan for Twi. We generate 12,500 synthetic two-speaker mixtures from LibriSpeech: $\mathbf{x}_{\text{mix}} = \mathbf{x}_{\text{target}} + \alpha \mathbf{x}_{\text{interferer}}$, varying $\text{SIR} \in \{0, 5\}$ dB and overlap ratio $\in \{0.1, 0.2, 0.3, 0.4, 0.5, 1.0\}$. Each mixture preserves the aligned clean target waveform for computing oracle masks during training. The recipes are split into 10,000 train, 1,250 validation, and 1,250 test samples, with speaker embeddings precomputed for all 2,588 unique enrollment utterances.

5 Numerical Experiments

5.1 Whisper fine-tuning for Twi

Whisper does not natively support Twi (Akan), producing near-100% WER and CER out of the box. We fine-tuned Whisper-small (244M params) on the WAXAL Akan dataset, reducing clean-speech WER from $\sim 100\%$ to 33.3% and CER from $\sim 100\%$ to 12.1%. This fine-tuned model serves as the Twi baseline for our overlap experiments.

5.2 Baseline: overlap degrades ASR across languages

We evaluated Whisper-tiny (39M params) on English and the fine-tuned Whisper-small on Twi across all SIR levels and overlap ratios.

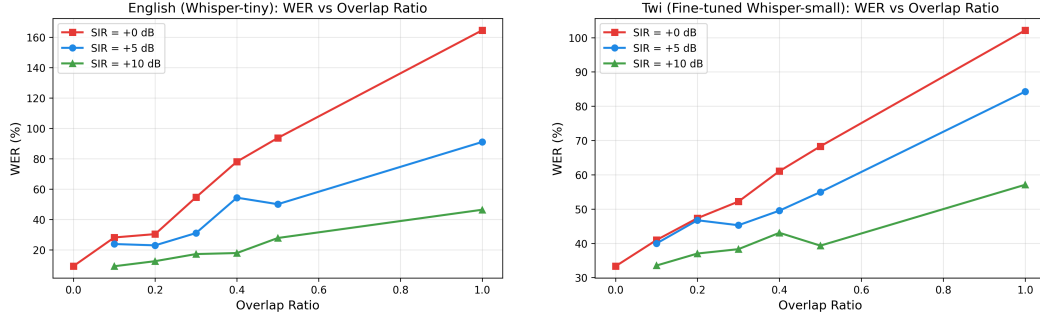


Figure 2: WER vs. overlap ratio for English (left, Whisper-tiny, clean baseline 8.1%) and Twi (right, fine-tuned Whisper-small, clean baseline 34.7%) at three SIR levels. Both languages show the same degradation pattern: WER increases with overlap ratio and decreases with SIR. At 0 dB SIR with full overlap, English WER exceeds 100%. This confirms the problem is language-agnostic.

5.3 Training and WER evaluation

We trained with batch size 16 (maximum that fits in GPU memory with 14-second random crops), initial learning rate 10^{-3} , Adam optimizer, hybrid loss $\alpha = 0.1$, power-law compression $p = 0.3$, and early stopping with patience 7. The learning rate is halved by ReduceLROnPlateau whenever validation loss does not improve for 3 consecutive epochs.

We evaluated two trained models on the 1,250-sample test set: the hybrid loss model ($\alpha = 0.1$) and the power-law reconstruction loss model ($\alpha = 0$). Both were trained until validation loss plateaued.

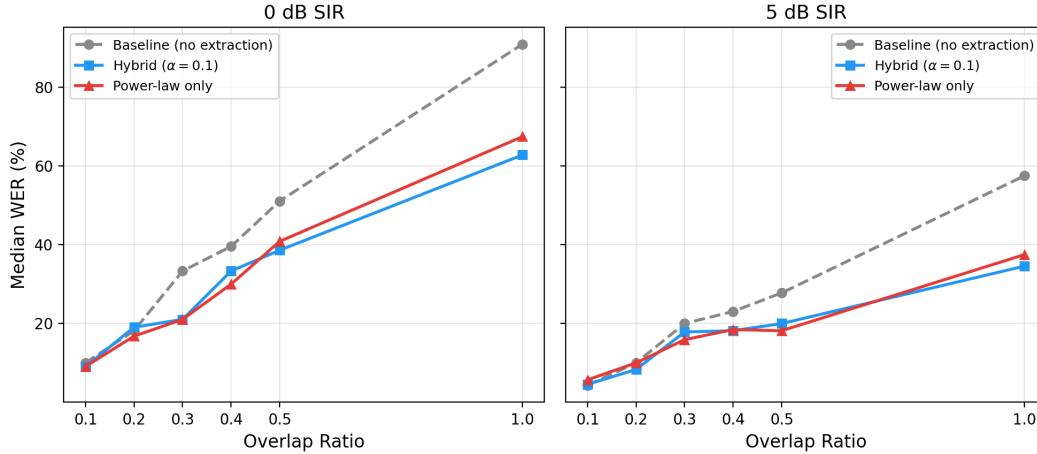


Figure 3: Median WER vs. overlap ratio for baseline (no extraction), hybrid loss ($\alpha = 0.1$), and power-law reconstruction loss only ($\alpha = 0$) at 0 dB SIR (left) and 5 dB SIR (right). Both extraction models reduce WER at overlap ratios ≥ 0.3 , with the largest gains at full overlap.

Both extraction models significantly reduce median WER compared to the baseline at overlap ratios ≥ 0.3 . At full overlap (1.0), the hybrid model reduces WER from 90.9% to 62.8% at 0 dB and from 57.5% to 34.6% at 5 dB. The power-law model achieves comparable gains (67.4% and 37.5% respectively). Neither model consistently dominates the other: hybrid is slightly better at very high overlap (1.0), while power-law is slightly better at medium overlap (0.3–0.5). At low overlap (≤ 0.2), extraction provides marginal or no benefit, as the interference is brief enough for Whisper to handle without assistance. We also verified that running extraction on clean speech (no interferer, overlap = 0) causes no noticeable WER degradation across all trained models, confirming the extractor does not suppress the target signal when no interference is present.

5.4 Effect of training distribution: hard cases only

We hypothesized that training on low-overlap mixtures wastes model capacity on easy cases where Whisper already performs well. To test this, we regenerated the dataset restricting overlap to $\{0.5, 1.0\}$ only and trained a power-law reconstruction model ($\alpha = 0$) on this harder distribution. Table 1 shows the median WER on the corresponding held-out test samples.

Table 1: Median WER (%) for the hard-distribution model (trained on overlap $\in \{0.5, 1.0\}$ only) vs. baseline Whisper. Δ is extraction minus baseline (negative = improvement).

SIR (dB)	Overlap	Baseline	Extracted	Δ
0	0.5	52.5	48.0	-4.5
0	1.0	92.3	93.8	+1.5
5	0.5	25.0	25.0	0.0
5	1.0	50.0	50.0	0.0

The hard-distribution model shows no meaningful improvement over baseline and even degrades slightly at full overlap (0 dB SIR). We additionally evaluated this model on the lower overlap conditions it was not trained on (overlap $\in \{0.1, 0.2, 0.3, 0.4\}$): improvements there are similarly negligible, with median WER changes of ≤ 5 percentage points at most conditions and near-zero change at SIR = 5 dB. This confirms the hard-distribution model failed to learn a general masking strategy across the board, not just at high overlap.

This is a worse result than the model trained on the full overlap distribution, which achieved reductions of up to 28 percentage points at full overlap. We attribute this to the loss of easy-case supervision: low-overlap examples provide clean, unambiguous oracle masks that anchor the model’s learning. Removing them leaves the model with only the hardest, most ambiguous training signal, leading to weaker generalization.

5.5 Curriculum learning

6 Discussion

The extraction network reduces WER most where baseline Whisper struggles most: at high overlap ratios and low SIR. At 0 dB SIR with full overlap, median WER drops from 90.9% to 62.8%, a 28 percentage point reduction. At 5 dB SIR with full overlap, the reduction is 57.5% to 34.6%. The difference between the hybrid loss ($\alpha = 0.1$) and pure power-law reconstruction loss ($\alpha = 0$) is minimal in terms of downstream WER, suggesting that the choice of training objective matters less than the overall extraction approach.

However, the extracted WER at full overlap (62.8% at 0 dB) remains far from the clean-speech baseline of 8.1%. The model has not fully converged during training: the validation loss plateaued well above the oracle mask floor, indicating that the predicted masks are still far from optimal. Longer training, higher model capacity, or architectural improvements such as skip connections could close this gap.

Our training data was uniformly distributed across overlap ratios (0.1, 0.2, 0.3, 0.4, 0.5, 1.0), meaning the model spent equal time on easy cases (low overlap, where Whisper already performs well) and hard cases (high overlap, where extraction is most needed). Biasing the training distribution toward higher overlap ratios, or using a curriculum that starts with easy examples and gradually increases difficulty, could improve performance on the conditions where extraction matters most.

We demonstrated that overlapping speech degrades ASR in both English and Twi, confirming the problem is language-agnostic. The extraction network operates on spectrograms and speaker embeddings rather than language-specific features, so it should in principle generalize across languages. We did not have time to evaluate the extraction network on Twi, but this is a natural next step: training on English and testing on Twi would reveal whether the learned masking strategy transfers cross-lingually without additional fine-tuning.

Table 2 summarises median WER across all seven trained configurations. *Curriculum Easy* was trained only on overlap $\in \{0.1, 0.2, 0.3\}$; *Curriculum Medium* resumes from that checkpoint and adds overlap $\{0.4, 0.5\}$ with upsampling weight $2\times$; *Curriculum Hard* resumes from the medium checkpoint and trains on the full overlap range with upsampling weights $[1, 1, 1, 2, 2, 3]$, progressively concentrating gradient signal on the hardest conditions.

Table 2: Median WER (%) for all models across SIR and overlap conditions. **Bold** indicates the best extraction model per row. *Curriculum Easy* was trained on overlap ≤ 0.3 only; *Curriculum Medium* extends to overlap ≤ 0.5 via warm-start fine-tuning; *Curriculum Hard* resumes from Medium and trains across all overlaps with weights $[1, 1, 1, 2, 2, 3]$.

SIR (dB)	Overlap	Baseline	Hard-dist. (PL, $\{0.5, 1.0\}$)	Power Law ($\alpha = 0$)	Hybrid ($\alpha = 0.1$)	Curriculum Easy	Curriculum Medium	Curriculum Hard
0	0.1	10.0	9.5	9.1	9.1	9.1	9.1	9.1
0	0.2	18.2	18.2	16.8	19.1	18.3	16.2	11.0
0	0.3	33.3	28.1	21.1	21.1	22.2	20.0	17.2
0	0.4	39.6	37.5	30.0	33.3	37.5	29.2	26.9
0	0.5	51.1	48.0	40.9	38.6	43.7	36.5	34.0
0	1.0	90.9	93.8	67.4	62.8	84.6	61.5	58.8
5	0.1	4.3	4.8	5.7	4.5	4.8	3.6	3.4
5	0.2	10.0	10.0	10.0	8.3	9.5	8.7	6.7
5	0.3	20.0	20.0	15.9	17.4	17.4	15.4	14.8
5	0.4	23.1	22.2	18.5	18.2	20.0	17.9	13.5
5	0.5	27.8	25.0	18.2	20.0	23.8	16.7	16.7
5	1.0	57.5	50.0	37.5	34.6	48.9	36.0	34.3

7 Conclusions

We built a target-speaker speech-to-text system that uses a speaker-conditioned masking network to isolate a target speaker from a two-speaker mixture before transcription with Whisper. The system reduces median WER by up to 28 percentage points at full overlap compared to running Whisper directly on the mixture. We showed that speaker overlap degrades ASR performance in both English and Twi, and that a lightweight extraction module (9.3M parameters) trained with a hybrid loss can partially recover the target speaker’s signal. We also found that restricting training to high-overlap cases only is counterproductive: the model relies on easy low-overlap examples to anchor its masking behavior, and removing them degrades performance.

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Use of AI

We used Claude (Anthropic) to assist with implementation, debugging, generating plots, and documenting the codebase, as well as to brainstorm ideas for improving model performance. All architecture decisions, experimental design, and analysis were done by the authors. AI-generated code was reviewed and tested before use.

Consent for Information Sharing

- All group members consent to allow the project materials (report, slides, and presentation recording) to be shared on the course page for future students.

Consent for Use of Project Information in Statistical Analysis

- All group members consent to allow anonymized information about the project (e.g., topic, methods, outcomes, grades) to be used by the instructor for statistical analysis and course improvement.

Group Identification

- Group number: 28
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